



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 3

Introduction to OOPS Programming

SESSION / ASSIGNMENT – 7

TASKS :

1. Create a text file called my_fav_songs.txt and write the names of your 5 favorite songs into it using ofstream.



```
#include <iostream>
#include <fstream>
using namespace std;

int main() {

    ofstream file("my_fav_songs.txt");

    file << "Blinding Lights" << endl;
    file << "Shape of You" << endl;
    file << "Perfect" << endl;
    file << "Believer" << endl;
    file << "Let Me Down Slowly" << endl;

    file.close();

    cout << "5 favorite songs written to the file successfully.";

    return 0;
}
```

OUTPUT:

```
5 favorite songs written to the file successfully.
```

```
≡ my_fav_songs.txt
1  Blinding Lights
2  Shape of You
3  Perfect
4  Believer
5  Let Me Down Slowly
6
```

2. Read all song names from my_fav_songs.txt using ifstream and display each song on a new line in the console.



```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    ifstream file("my_fav_songs.txt");

    string song;

    while (getline(file, song)) {
        cout << song << endl;
    }

    file.close();

    return 0;
}
```

OUTPUT:

```
Blinding Lights
Shape of You
Perfect
Believer
Let Me Down Slowly
```

3. Modify your code to append a new song name entered by the user to my_fav_songs.txt without overwriting the existing list.

Hint: Open the file in append mode using ofstream.



```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    string song;

    cout << "Enter a new song name: ";
    getline(cin, song);

    // Open file in append mode
    ofstream file("my_fav_songs.txt", ios::app);

    file << song << endl;

    file.close();

    cout << "Song added successfully.";

    return 0;
}
```

OUTPUT:

```
Enter a new song name: Tunak Tunak
Song added successfully.
```

```
≡ my_fav_songs.txt
1  Blinding Lights
2  Shape of You
3  Perfect
4  Believer
5  Let Me Down Slowly
6  Tunak Tunak
7
```

4. Build a Flipkart-style wishlist tracker: ask the user to enter 3 product names and prices, save them to a file called wishlist.txt, then read the file and display each product with its price.



```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    string product;
    double price;

    // Open file for writing
    ofstream outFile("wishlist.txt");

    // Enter 3 products
    for (int i = 1; i <= 3; i++) {

        cout << "Enter product " << i << " name: ";
        getline(cin, product);

        cout << "Enter price: ";
        cin >> price;
        cin.ignore();

        outFile << product << " " << price << endl;
    }

    outFile.close();

    // Open file for reading
    ifstream inFile("wishlist.txt");

    cout << endl;
    cout << "----- Wishlist -----" << endl;

    while (inFile >> product >> price) {
        cout << "Product: " << product
            << " | Price: Rs. " << price << endl;
    }

    inFile.close();

    return 0;
}
```

OUTPUT:

```
Enter product 1 name: laptop
Enter price: 100000
Enter product 2 name: keyboard
Enter price: 199
Enter product 3 name: mouse
Enter price: 300

----- Wishlist -----
Product: laptop | Price: Rs. 100000
Product: keyboard | Price: Rs. 199
Product: mouse | Price: Rs. 300
```

```
≡ wishlist.txt
1  laptop 100000
2  keyboard 199
3  mouse 300
4
```

5. Given a file called `insta_followers.txt` containing Instagram usernames (one per line), write a program to count and display the total number of followers listed in the file.

Constraint: Do not use any array or vector to store the names—just count as you read.



```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    ifstream file("insta_followers.txt");

    string username;
    int count = 0;

    while (getline(file, username)) {
        count++;
    }

    file.close();

    cout << "Total Followers: " << count << endl;

    return 0;
}
```

File - insta_followers.txt

```
≡ insta_followers.txt
1  haaris
2  john
3  alex
4  sarah
5  mike
```

OUTPUT:

```
Total Followers: 5
```